SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	Sasikumar Megha	**Reg. No.:**	192471038
Section / Batch:		**Date:**	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:		**100 Marks**

SECTION A – CONCEPT MAPPING

48/50

Code of Conduct

I Sasikumar Megha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Megha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1	24	24	20	20	10	48
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature:	Signature:	Signature:
Date: 20/7/26	Date:	Date:

: Sasikumar Meghe
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CSA0601

Q1) Consider the cocept map: Algorithm → Input size (n) → Time complexity → Growth rate → Asymptopic Notation

Explain how the increase in input size affects the growth rate and how this lead to classification of time complexity. Extend the map by including Big-O, Big-Teta, and Big-Omega as branch under asymptomic notation. Justify how much each notation provides a different perespective of algorithm performance.

Algorithm

↓

Inputsize (n)

Effect of increasing Input size

* As n increases, number of operations increases.
* Growth rate shows how fast T(n) grows
* Helps classify algorithms by efficiency.

↓

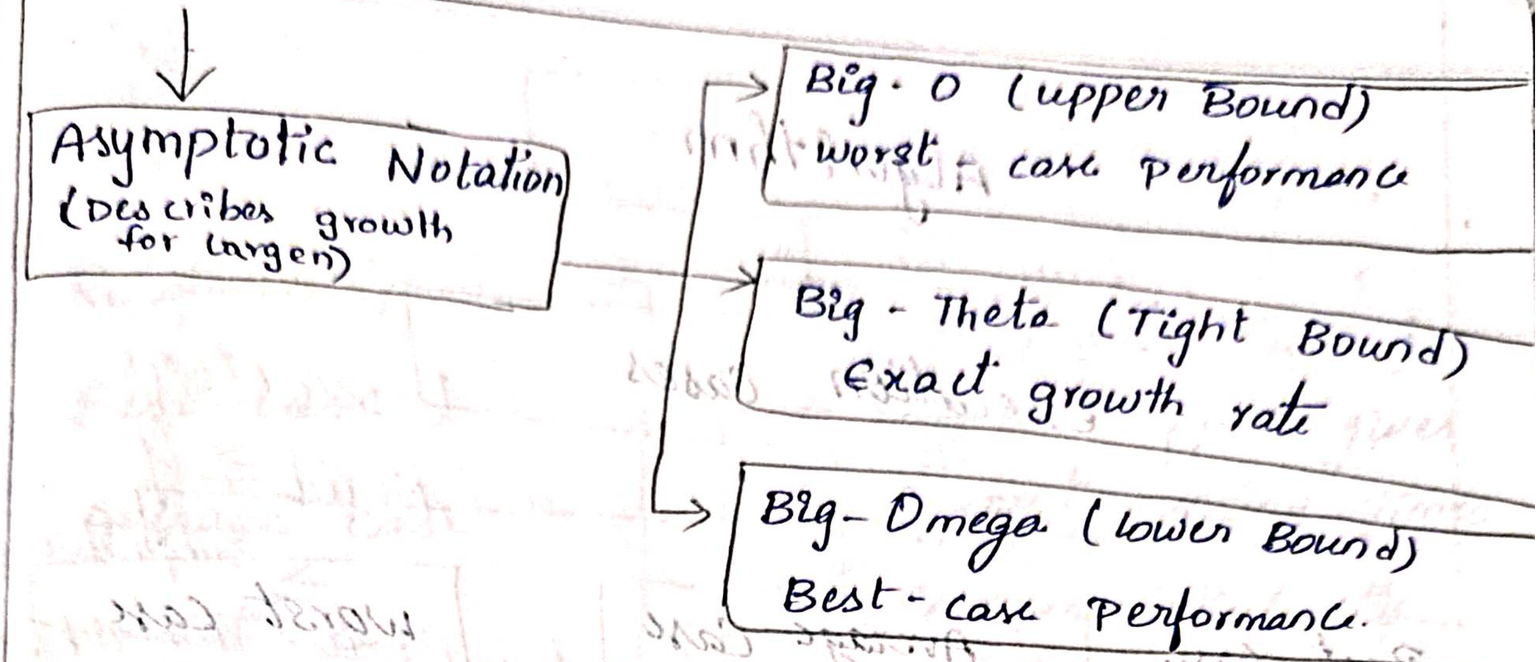
Time complexity (T(n))

Common Growth Rates

$O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(n^3) < O(2^n)$

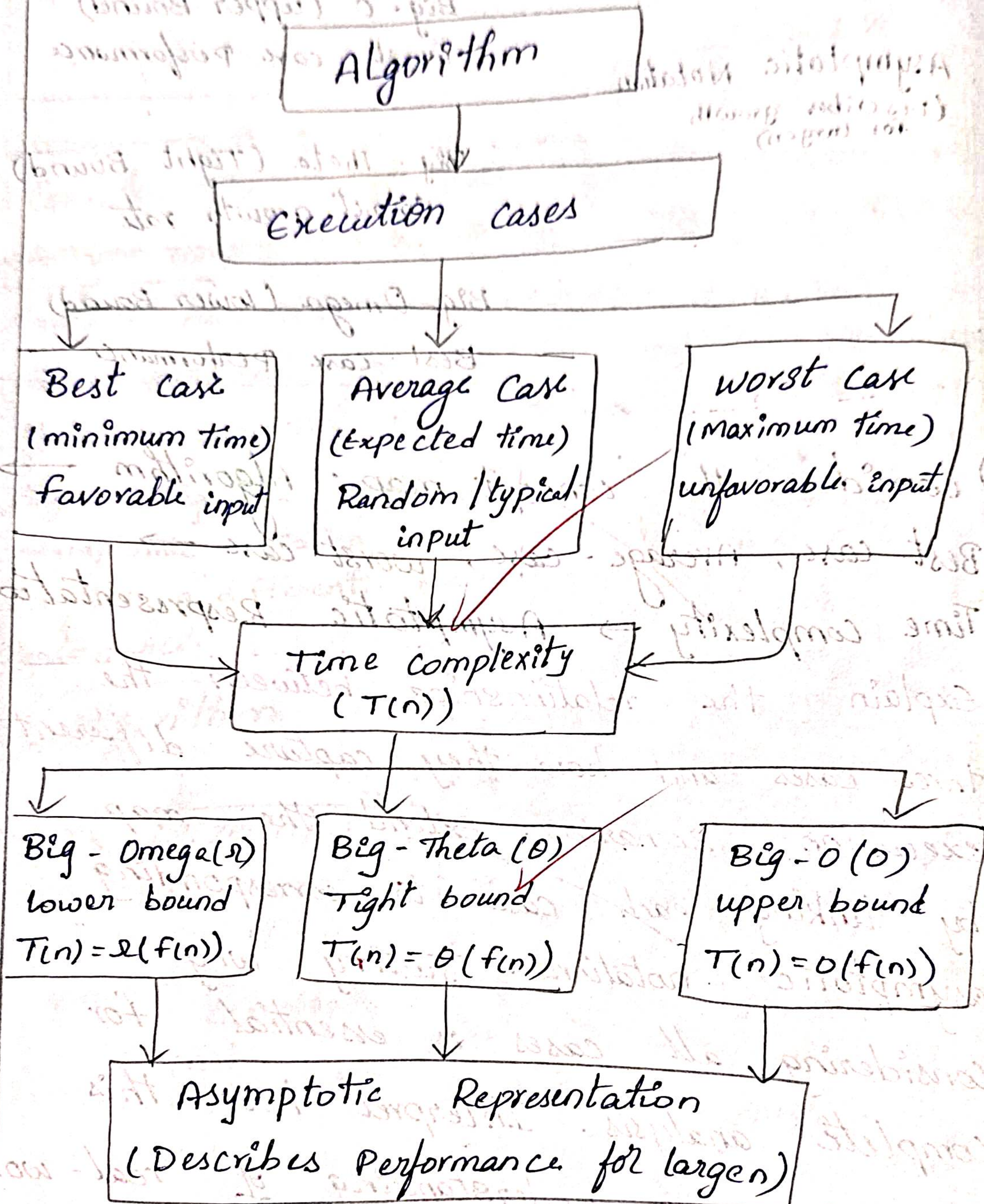
↓

Growth rate (rate at which T(n) increase with n)



Q2) Consider the concept map: Algorithm → Best-case, Average-case, worst case → Time complexity → Asymptotic Respresentation

Explain the relationships between the three cases and how they capture different execution scenarios. Extend the map by linking each case to corresponding asymptotic notations. Justify why considering all cases is essential for complete analysis. Interpret how this map improves understanding of real-world algorithms performance.


Algorithm
Execution cases
Best case (minimum time) favorable input
Average case (Expected time) Random / typical input
worst case (Maximum time) unfavorable input
Time complexity (T(n))
Big - Omega(Ω) lower bound T(n) = Ω(f(n))
Big - Theta (θ) Tight bound T(n) = θ(f(n))
Big - O (O) upper bound T(n) = O(f(n))
Asymptotic Representation (Describes Performance for large n)

Relationship

Best case
$\leq$
Average case
$\leq$
worst case

why all cases?

* Different input gives different running times.
* Ensures reliability
* Helps in algorithm comparsion.
* Important for real-world application.

Q13) Consider the concept map: Divide and conquer → Recursive calls → Recurrence Relation → Master Theorem → Complexity Analysis. Explain how recursive calls from recurrence relations in divide-and-conquer algorithms. Extend the map by including parameters a, b and f(n). Justify how the Master Theorem simplifies solving such recurrencs. Interpet how this structured mapping support efficient analysis.

Divide and conquer
(Break problem into smaller subproblems)

↓

Recursive calls
(solve subproblems recursively)

↓

Recurrence Relation

$T(n) = aT(n/b) + f(n)$

↓

Master Theorem
(uses a, b and f(n))

↓

Complexity Analysis
(Determine T(n))

Parameters

a : Number of subproblems

b : Factor by which input size is reduced

f(n) : work done outside recursive calls.

Master Theorem Cases

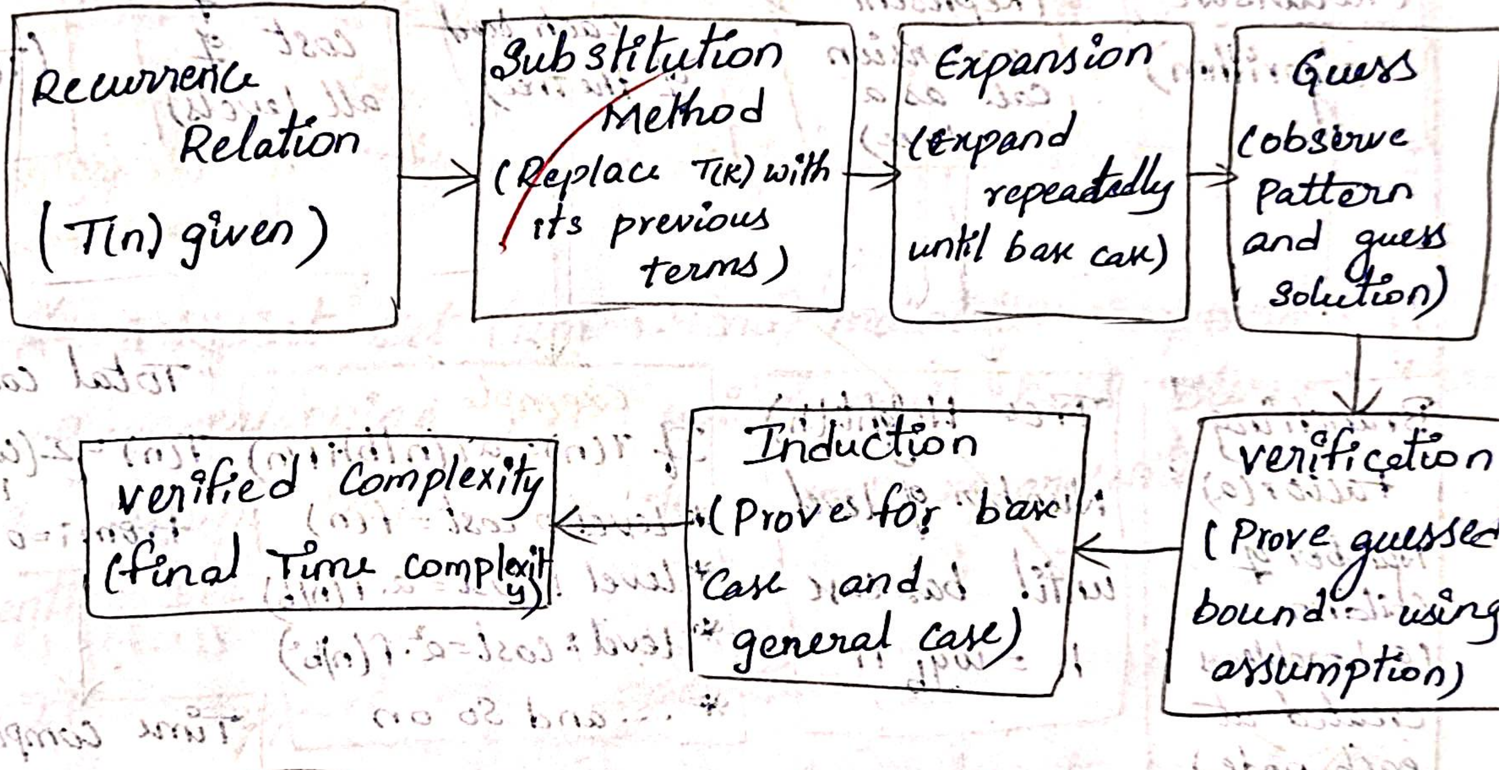
1) If $f(n) = O(n^{\log_b a - \epsilon})$, $T(n) = \Theta(n^{\log_b a})$

2) If $f(n) = \Theta(n^{\log_b a})$, $T(n) = \Theta(n^{\log_b a} \log n)$

3) If $f(n) = \Omega(n^{\log_b a + \epsilon})$ and regularity holds, $T(n) = \Theta(f(n))$

Q 94) Consider the concept map : Recurrence Relation → Substitution Method → Expansion → Induction → verified Complexity.

Explain how substitution method solves recurrence relation through expansion. Extend the map by adding Guess and verification nodes. Jusify why induction is required to confirm correctness. Interpret how this map ensures systematic and accurate complexity.



Why Induction ?

* Expansion shows pattern but not proof.
* Induction confirms that the gussed bound holds for all $n \geq n_0$.
* Ensures correctness of the derived complexity

Q 95). Consider the concept map: Recursion → Recursion Tree → level-wise cost → Total cost → Time complexity. Explain how recursion can be represented using a tree structure. Extend the map by including tree height and branching factor. Justify how summation of level costs gives total complexity. Interpret how this mapping simplies recursive analysis.